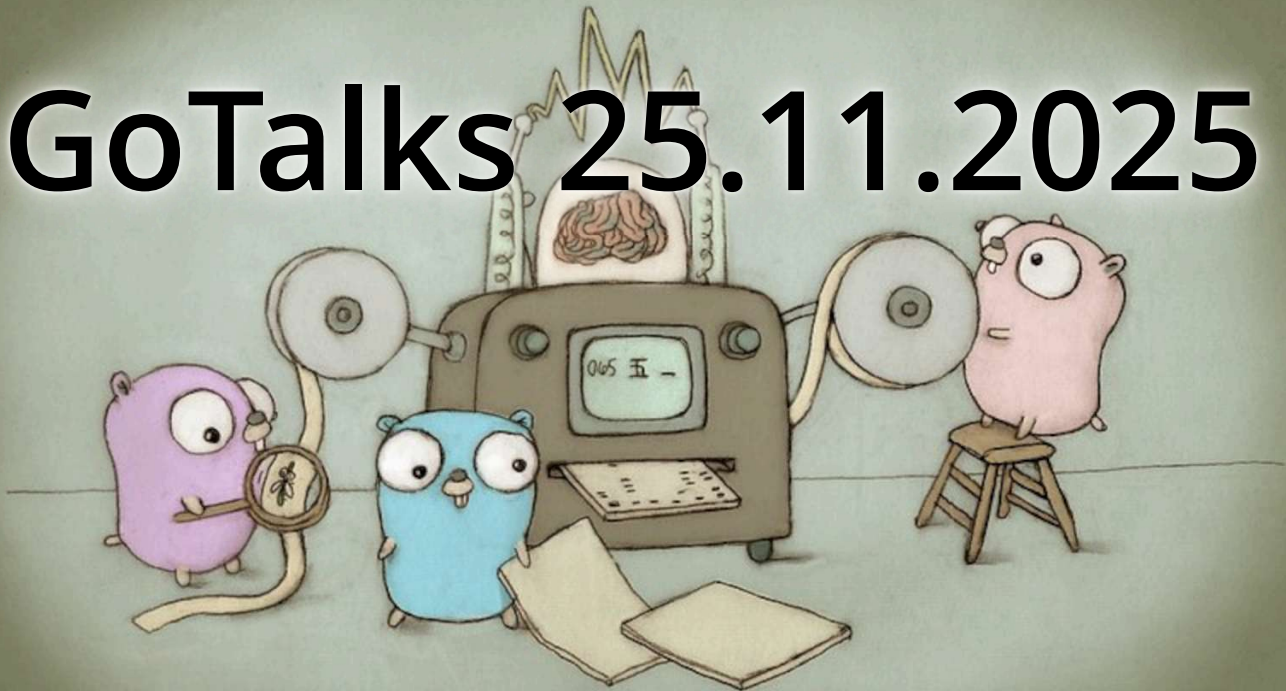


GoTalks 25.11.2025



How to reach us



meetup.com/Golang-ZG



[@golangzg](https://www.youtube.com/@golangzg)



github.com/golanghr/golangzg



[@golangzg.bsky.social](https://bsky.app/@golangzg.bsky.social)



invite.slack.golangbridge.org



- golangci-lint
 - [golangci-lint.run](#)
 - Fast Go linters runner
 - Supports multiple linters
 - Highly configurable
 - configuration file: `.golangci.yml`
 - memory inefficient
 - installations is not native `go install`
 - each new go version does not work until golangci-lint releases a new version

- staticcheck
 - staticcheck.io
 - A state-of-the-art linter for Go
 - Focuses on code correctness, performance, and style issues
 - Can be used standalone or integrated into other tools
 - Regularly updated with new checks and improvements
 - installation: `go install honnef.co/go/tools/cmd/staticcheck@latest`

- revive
 - revive.run
 - Fast, configurable, extensible linter for Go
 - Focuses on code style and best practices
 - Highly configurable with custom rules
 - installation: `go install`
`github.com/mgechev/revive@latest`
 - does not have issues with memory like golangci-lint
 - does not have issue with new go versions

- betteralign
 - github.com/dkorunic/betteralign
 - Make your Go programs use less memory (maybe)
 - Aligns struct fields to minimize memory usage due to padding
 - installation: `go install github.com/dkorunic/betteralign@latest`
 - usage: `betteralign ./...`
 - automatic modifications: `betteralign -apply ./...`
 - uses DST and not AST

- gofumpt
 - github.com/mvdan/gofumpt
 - A stricter gofmt
- modernize
 - `go run`
`golang.org/x/tools/gopls/internal/analysis/modernize/c`
`modernize@latest -fix -test ./...`
- betteralign
 - also does formatting fixes

- go-licenses
 - github.com/google/go-licenses
 - Analyze and display the licenses of Go packages
 - installation: `go install github.com/google/go-licenses/v2@latest`
 - `go-licenses report --ignore "$PROJECT" \`
`--ignore "github.com/golang/freetype/raster" .`
 - check allowed licenses:

```
go-licenses check --allowed_licenses="$ALLOWED_LICENSES"  
--ignore "$PROJECT" \  
--ignore "github.com/oktalz/present" .
```



- govulncheck
 - pkg.go.dev/golang.org/x/vuln/cmd/govulncheck
 - Check your Go programs for known vulnerabilities
 - installation: `go install`
`golang.org/x/vuln/cmd/govulncheck@latest`
 - usage:
 - `govulncheck ./...`
 - `govulncheck -show verbose ./...`

- gotestsum
 - github.com/gotestyourself/gotestsum
 - A test runner that generates JUnit XML and more readable output
 - installation: `go install github.com/gotestyourself/gotestsum@latest`
 - `gotestsum --hide-summary=output --format-hide-empty-pkg -- ./...`

`github.com/oktalz/present/parsing:`

- ✓ Find data (0.00s)
- ✓ Find data use **case 1** (0.00s)
- ✓ Find data use **case 2** (0.00s)



- yq
 - github.com/mikefarah/yq
 - Command-line YAML processor
 - installation: `go install`
`github.com/mikefarah/yq/v4@latest`
 - can work with YAML, JSON, XML, CSV, and TOML files

- mlr
 - github.com/johnkerl/miller
 - A command-line tool for processing CSV, TSV, and JSON files
 - installation: `go install`
`github.com/johnkerl/miller/v6/cmd/mlr@latest`
 - provides powerful data manipulation capabilities similar to awk, sed, and cut

- taskfile
 - <https://taskfile.dev/>
 - A simple task runner / build tool that aims to be simpler and easier to use than Make
 - installation: `go install github.com/go-task/task/v3/cmd/task@latest`
 - usage: create a `taskfile.yml` and run `task <task-name>`

Development Utilities

- air
 - <https://github.com/air-verse/air>
 - Live reload for Go applications
 - installation: `go install github.com/air-verse/air@latest`
 - usage:
 - run `air` in the project directory
 - `air --build.cmd "go build -o bin/api cmd/run.go" --build.bin "./bin/api"`
 - automatically rebuilds and restarts the application on file changes



Generation of code

- go-method-gen
 - github.com/haproxytech/go-method-gen
 - A tool to generate boilerplate code for Go methods
 - installation: `go install github.com/haproxytech/go-method-gen@latest`
 - generates Equal and Diff methods for structs (Merge, Clone is planned)
 - allows customization



Docker & Kubernetes

- lazydocker
 - A simple terminal UI for both docker and docker-compose, written in Go
 - installation: `go install github.com/jesseduffield/lazydocker@latest`
 - provides an easy way to manage Docker containers, images, volumes, and networks from the terminal
 - supports viewing logs, stats, and executing commands within containers





Docker & Kubernetes

- k9s
 - A terminal UI to interact with your Kubernetes clusters
 - installation: `go install github.com/derailed/k9s@latest`
 - provides a terminal-based interface for managing and monitoring Kubernetes clusters
 - supports various Kubernetes resources and operations



Docker & Kubernetes

- kind
 - kind.sigs.k8s.io
 - Kubernetes IN Docker - a tool for running local Kubernetes clusters using Docker container nodes
 - installation: `go install sigs.k8s.io/kind@latest`
 - useful for testing Kubernetes applications locally
 - supports multi-node clusters



- k0s
 - k0sproject.io
 - `curl -sSf https://get.k0s.sh | sudo sh`
 - A zero-friction Kubernetes distribution
 - provides a simple way to set up and manage Kubernetes clusters
 - supports single-node and multi-node deployments

Go Obfuscation

- garble
 - github.com/burrowers/garble
 - A tool to obfuscate Go binaries
 - installation: `go install mvdan.cc/garble@latest`
 - usage: `garble build` instead of `go build`
 - obfuscates symbol names and strings in the binary to make reverse engineering more difficult

- act
 - github.com/nektos/act
 - installation: `go install github.com/nektos/act@latest`
 - A tool to run GitHub Actions locally
 - allows testing and debugging GitHub Actions workflows without pushing to GitHub
 - supports most GitHub Actions features and syntax
 - usage: `act -j <job-name>`

- goreleaser
 - goreleaser.com
 - A tool to automate the release process of Go projects
 - installation: `go install github.com/goreleaser/goreleaser/v2@latest`
 - usage:
 - `goreleaser release --snapshot --clean`
 - `goreleaser release`
 - supports building, packaging, and publishing Go binaries to various platforms and package managers
 - Works with Go, Rust, Zig, Bun, Deno

Go Integrated Tools

- gopls
 - pkg.go.dev/golang.org/x/tools/gopls
 - The official Go language server
 - Provides IDE features like auto-completion, go-to-definition, and refactoring
 - installation: `go install golang.org/x/tools/gopls@latest`
 - usually integrated into editors like VSCode, Vim, etc.



Go Integrated Tools

- delve
 - github.com/go-delve/delve
 - A powerful debugger for Go programs
 - installation: `go install github.com/go-delve/delve/cmd/dlv@latest`
 - supports various debugging modes: local, remote, headless
 - can be integrated with IDEs for a better debugging experience



Go Integrated Tools

- pprof
 - A tool for profiling Go programs
 - Built into the standard library
 - usage: import `_ "net/http/pprof"` and start an HTTP server
 - access profiles via `http://localhost:6060/debug/pprof/`
 - can generate CPU, memory, goroutine, and block profiles
 - visualize profiles using `go tool pprof`



Go Integrated Tools

- `go test`
 - A built-in tool for measuring code coverage in Go tests
 - usage: `go test -coverprofile=coverage.out ./...`
 - view coverage report using `go tool cover -html=coverage.out`
 - helps identify untested parts of the codebase

Go Integrated Tools

- `go doc -http`
 - A built-in tool to serve Go documentation locally
 - usage: `go doc -http`
 - provides a web interface for browsing Go package documentation

Shameless self-promotion

- present
 - github.com/oktalz/present
 - A Go application for creating presentations
 - installation: `go install`
`github.com/oktalz/present@latest`
 - usage: `present` command to serve presentations locally
 - supports markdown-like syntax for slides
 - highly customizable with CSS and HTML



Thank you for your
attention! 😊